**UNIVERSITY OF CALIFORNIA, SAN DIEGO**
**Electrical & Computer Engineering Department**
**ECE 101 - Fall 2023**
*Linear Systems Fundamentals*

# FINAL EXAM

**You are allowed two 2-sided sheets of notes.**
**Tables 3.1, 3.2, 4.1, 4.2, 5.1, 5.2, 9.1, and 9.2 are attached.**
**No books, no other notes, no calculators.**

PRINT YOUR NAME _____

Signature _____

Student ID Number _____

**Your signature confirms that you have completed this exam in accordance with the ECE 101 Academic Integrity Agreement.**

| Problem | Weight | Score |
|---|---|---|
| 1 | 20 | |
| 2 | 20 | |
| 3 | 20 | |
| 4 | 20 | |
| 5 | 20 | |
| 6 | 20 | |
| Total | 120 | |

**Please do not begin until told.**
**Show your work.**
**Use back of previous page and attached scratch sheets as needed.**
**Scan, upload, and tag your solutions on Gradescope at the end of the exam.**

**Good luck! You've got this!**

**Name/Student ID:** _____

**Problem 1 [CTFT Properties]** (20 points, 5 points each part)

Let $x(t) = u(t+3) - u(t-1)$. Let $X(j\omega)$ denote its Fourier transform. In the following problems, refer to the attached tables, as necessary, to explain your solutions.

(a) Determine $X(j\omega)$.

(b) Determine $\int_{-\infty}^{\infty} X(j\omega)d\omega$.

(c) Let $y(t) = x(\frac{t}{2})$, with Fourier transform $Y(j\omega)$. Determine $Y(j0)$.

(d) Determine $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$

**Write answers to parts (a) through (d) on the following pages.**

**Justify your answers.**

**Name/Student ID:** _____

**Problem 1 [CTFT Properties] (cont.)**

Let $x(t) = u(t+3) - u(t-1)$. Let $X(j\omega)$ denote its Fourier transform.

(a) Determine $X(j\omega)$.

**Name/Student ID:** _____

**Problem 1 [CTFT Properties] (cont.)**

Let $x(t) = u(t+3) - u(t-1)$. Let $X(j\omega)$ denote its Fourier transform.

(b) Determine $\int_{-\infty}^{\infty} X(j\omega) d\omega$.

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**Problem 1 [CTFT Properties] (cont.)**

Let $x(t) = u(t+3) - u(t-1)$. Let $X(j\omega)$ denote its Fourier transform.

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**Problem 1 [CTFT Properties] (cont.)**

Let $x(t) = u(t+3) - u(t-1)$. Let $X(j\omega)$ denote its Fourier transform.

(d) Determine $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$

**Name/Student ID:** _____

**Problem 2 [DTFT]** (20 points, 5 points each part)

This problem deals with the DT LTI system $S$ with impulse response

$$h[n] = \left(\frac{1}{2}\right)^{-n} u[-n] + \left(\frac{1}{2}\right)^{n} u[n].$$

(a) Precisely sketch the signal $h[n]$ in the time interval $[-2, 2]$.

(b) Let $x[n] = u[n-1]$ be the input signal to the system $S$.
Determine the value of the corresponding output signal $y[n]$ at $n = 0$.

(c) Determine the frequency response $H(e^{j\omega})$ of the system $S$.
Express it in a form that does not use any exponential functions of $\omega$.

(d) Is the system $S$ invertible? If not, why not? If so, determine the frequency response $G(e^{j\omega})$ of its inverse system.

**Write answers to parts (a) through (d) on the following pages.**

**Justify your answers.**

**Name/Student ID:** _____

**Problem 2 [DTFT] (cont.)**

This problem deals with the DT LTI system $S$ with impulse response

$$h[n] = \left(\frac{1}{2}\right)^{-n} u[-n] + \left(\frac{1}{2}\right)^{n} u[n].$$

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(b) Let $x[n] = u[n-1]$ be the input signal to the system $S$.
Determine the value of the corresponding output signal $y[n]$ at $n = 0$.

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(d) Is the system $S$ invertible? If not, why not? If so, determine the frequency response $G(e^{j\omega})$ of its inverse system.

**Name/Student ID:** _____

**Problem 3 [Amplitude Modulation]** (20 points, 5 points each part)

Consider the signals $x_1(t) = \frac{\sin(10t)}{\pi t}$ and $x_2(t) = \frac{\sin(5t)}{\pi t}$.

(a) Determine the Fourier transforms $X_1(j\omega)$ and $X_2(j\omega)$. Sketch them precisely.

(b) Let signal $y(t) = y_1(t) + y_2(t)$, where $y_1(t) = x_1(t)\cos(20t)$ and $y_2(t) = x_2(t)\cos(40t)$. Express the Fourier transform $Y(j\omega)$ of $y(t)$ in terms of $X_1(j\omega)$ and $X_2(j\omega)$. Sketch $Y(j\omega)$ precisely.

(c) The signal $y(t)$ is demodulated using multiplication by $\cos(20t)$ to form the signal $z(t) = y(t)\cos(20t)$. Express the Fourier transform $Z(j\omega)$ in terms of $Y_1(j\omega)$ and $Y_2(j\omega)$. Then express $Z(j\omega)$ in terms of $X_1(j\omega)$ and $X_2(j\omega)$. Sketch precisely the portion of $Z(j\omega)$ in the frequency range $[-50, 50]$.

(d) (i) Describe precisely a scheme to recover $x_1(t)$ from $z(t)$.

(ii) Describe a scheme to recover $x_2(t)$ from the signal $y(t)$.

**Write answers to parts (a) through (d) on the following pages.**

**Justify your answers.**

**Name/Student ID:** _____

**Problem 3 [Amplitude Modulation] (cont.)**

Consider the signals $x_1(t) = \frac{\sin(10t)}{\pi t}$ and $x_2(t) = \frac{\sin(5t)}{\pi t}$.

(a) Determine the Fourier transforms $X_1(j\omega)$ and $X_2(j\omega)$. Sketch them precisely.

**Name/Student ID:** _____

**Problem 3 [Amplitude Modulation] (cont.)**

(b) Let signal $y(t) = y_1(t) + y_2(t)$, where $y_1(t) = x_1(t)\cos(20t)$ and $y_2(t) = x_2(t)\cos(40t)$. Express the Fourier transform $Y(j\omega)$ of $y(t)$ in terms of $X_1(j\omega)$ and $X_2(j\omega)$. Sketch $Y(j\omega)$ precisely.

**Name/Student ID:** _____

**Problem 3 [Amplitude Modulation] (cont.)**

(c) The signal $y(t)$ is demodulated using multiplication by $\cos(20t)$ to form the signal $z(t) = y(t)\cos(20t)$. Express the Fourier transform $Z(j\omega)$ in terms of $Y_1(j\omega)$ and $Y_2(j\omega)$. Then express $Z(j\omega)$ in terms of $X_1(j\omega)$ and $X_2(j\omega)$. Sketch precisely the portion of $Z(j\omega)$ in the frequency range $[-50, 50]$.

**Name/Student ID:** _____

**Problem 3 [Amplitude Modulation] (cont.)**

(d) (i) Describe precisely a scheme to recover $x_1(t)$ from $z(t)$.

(ii) Describe a scheme to recover $x_2(t)$ from the signal $y(t)$.

**Name/Student ID:** _____

**Problem 4 [Sampling Theory]** (20 points, 5 points each part)

For each of the following signals, determine if a sampling period of $T = \frac{1}{4}$ seconds is sufficiently small to avoid aliasing and allow reconstruction of the signal from its samples. Justify your answers by explicit reference to the Fourier transforms of the signals.

(a) $x(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & |t| > 1/2 \end{cases}$

(b) $x(t) = \cos(\frac{3}{2}\pi t)e^{-j\frac{3}{2}\pi t}$

(c) $x(t) = \frac{\sin(3\pi t)}{\pi t} * \frac{\sin(6\pi t)}{\pi t}$

(d) $x(t) = e^{-3\pi t}u(t)$

**Write answers to parts (a) through (d) on the following pages.**

**Justify your answers.**

**Name/Student ID:** _____

**Problem 4 [Sampling Theory] (cont.)**

For each of the following signals, determine if a sampling period of $T = \frac{1}{4}$ seconds is sufficiently small to avoid aliasing and allow reconstruction of the signal from its samples. Justify your answers by explicit reference to the Fourier transforms of the signals.

(a) $x(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & |t| > 1/2 \end{cases}$

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For each of the following signals, determine if a sampling period of $T = \frac{1}{4}$ seconds is sufficiently small to avoid aliasing and allow reconstruction of the signal from its samples. Justify your answers by explicit reference to the Fourier transforms of the signals.

(b) $x(t) = \cos(\frac{3}{2}\pi t)e^{-j\frac{3}{2}\pi t}$

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**Problem 4 [Sampling Theory] (cont.)**

For each of the following signals, determine if a sampling period of $T = \frac{1}{4}$ seconds is sufficiently small to avoid aliasing and allow reconstruction of the signal from its samples. Justify your answers by explicit reference to the Fourier transforms of the signals.

(c) $x(t) = \frac{\sin(3\pi t)}{\pi t} * \frac{\sin(6\pi t)}{\pi t}$

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For each of the following signals, determine if a sampling period of $T = \frac{1}{4}$ seconds is sufficiently small to avoid aliasing and allow reconstruction of the signal from its samples. Justify your answers by explicit reference to the Fourier transforms of the signals.

(d) $x(t) = e^{-3\pi t}u(t)$

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**Problem 5 [Laplace Transform and LTI Systems]** (20 points)

The system function of a stable LTI system $S$ is

$$H(s) = \frac{s}{s^2 + 2s + 2}$$

(a) Determine the poles and zeros and sketch precisely the pole-zero plot.

(b) Let $H(j\omega)$ be the frequency response of the system $S$. Determine the magnitude $|H(j\omega)|$ at $\omega = 0, 1$, and in the limit as $\omega \to \pm\infty$.

(c) Sketch $|H(j\omega)|$ approximately. Is the system low-pass, high-pass, band-pass, or all-pass in nature?

**Write answers to parts (a) through (c) on the following pages. Justify your answers.**

**Name/Student ID:** _____

**Problem 5 [Laplace Transform and LTI Systems] (cont.)**

The system function of a stable LTI system $S$ is

$$H(s) = \frac{s}{s^2 + 2s + 2}$$

(a) (8 points)
Determine the poles and zeros and sketch precisely the pole-zero plot.

**Name/Student ID:** _____

**Problem 5 [Laplace Transform and LTI Systems] (cont.)**

The system function of a stable LTI system $S$ is

$$H(s) = \frac{s}{s^2 + 2s + 2}$$

(b) (8 points)
Let $H(j\omega)$ be the frequency response of the system $S$. Determine the magnitude $|H(j\omega)|$ at $\omega = 0, 1$, and in the limit as $\omega \to \pm\infty$.

**Name/Student ID:** _____

**Problem 5 [Laplace Transform and LTI Systems] (cont.)**

The system function of a stable LTI system $S$ is

$$H(s) = \frac{s}{s^2 + 2s + 2}$$

(c) (4 points)
Sketch $|H(j\omega)|$ approximately. Is the system low-pass, high-pass, band-pass, or all-pass in nature?

**Name/Student ID:** _____

**Problem 6 [Laplace Transform and Differential Equations]** (20 points)

Consider an LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}.$$

(a) Determine the system function $H(s)$ and the possible regions of convergence (ROC). For each region of convergence, determine whether the corresponding system is causal or stable. [Hint: Your answer to Problem 5(a) will be helpful here.]

(b) Determine the impulse response $h(t)$ when the system is causal. Express your answer in the form $Ce^{-\alpha t}\cos(\omega_0 t)u(t) + De^{-\alpha t}\sin(\omega_0 t)u(t)$, where $C, D, \alpha, \omega_0$ are constants. Use partial fraction expansion, and verify your answer using Table 9.2.

(c) Carefully sketch the canonical (direct-form) block diagram representing the causal LTI system in part (b).

**Write answers to parts (a) through (c) on the following pages.**

**Justify your answers.**

**Name/Student ID:** _____

**Problem 6 [Laplace Transform and Differential Equations] (cont.)**

Consider an LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}.$$

(a) (8 points)
Determine the system function $H(s)$ and the possible regions of convergence (ROC). For each region of convergence, determine whether the corresponding system is causal or stable. [Hint: Your answer to Problem 5(a) will be helpful here.]

**Name/Student ID:** _____

**Problem 6 [Laplace Transform and Differential Equations] (cont.)**

Let $S$ be the causal LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}.$$

(b) (8 points)
Determine the impulse response $h(t)$ when the system is causal. Express your answer in the form $Ce^{-\alpha t}\cos(\omega_0 t)u(t) + De^{-\alpha t}\sin(\omega_0 t)u(t)$, where $C, D, \alpha, \omega_0$ are constants. Use partial fraction expansion, and verify your answer using Table 9.2.

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**Problem 6 [Laplace Transform and Differential Equations] (cont.)**

Consider an LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}.$$

(c) (4 points)
Carefully sketch the canonical (direct-form) block diagram representing the causal LTI system in part (b).